H2O.ai: Learn to Use the KMeans Algorithm

Consumer trends and choices play a key role in enterprise strategies. Analysing huge volumes of customer data is not feasible manually. Hence, enterprises lean heavily on machine learning. H2O.ai is open source software for Big Data analysis. This article discusses the implementation of the KMeans algorithm with H2O.ai.

Most enterprises are taking a lot of interest in machine learning (ML) techniques, which help us use historical data to make better business decisions. ML algorithms discover patterns in data and construct mathematical models based on these discoveries. Such models can then be used to predict the occurrence of events of significance at some future date. As an example, if we have information about our customers (like name, age, address, financial position, marital status, etc), it would be possible to group them. Based on this grouping, it would be possible to gauge the probability of the identified group purchasing the new products that an enterprise may be ready to launch.

For any ML technique to be successful, it needs quality data and that too in volume. Till the arrival of Big Data technologies that allowed enterprises to crunch huge volumes of data in a distributed manner using commodity hardware, most machine learning techniques used sample data sets. These sets allowed ML algorithms to generate reasonably accurate results. But, as is well known, an inference is only as good as the data from which it is derived. Thus, it is preferable to derive inferences from the complete data set, instead of relying on sample data.

Using Big Data tools like Hadoop, the open source community created the Apache Mahout ML library. This library allows the execution of ML algorithms across huge volumes of data using commodity hardware. As enterprises grew more comfortable with and realised the importance of ML, they demanded high performance algorithms that would be able to process voluminous data, but also to do so at speed. This is one of the key drivers for applications like H2O.ai, developed by 0xdata (pronounced 'hexadata').

What is H2O?

H2O is an open source, in-memory application for machine learning. It does in-memory analytics on clusters, with distributed parallelised machine learning algorithms. By using in-memory data, it is able to perform compute-intensive operations at speed and deliver results much faster than other popular tools like Apache Mahout. H2O can be deployed as a standalone application on the Hadoop platform and on Amazon EC2. For storing data, it uses the main memory of the server on which it executes. In case of Hadoop, it can use HDFS as a data store. For the Amazon space, it can use Amazon S3.